

Automated Speech-to-Text Captioning for Videos and Noise Robustness Analysis Using OpenAI Whisper: A Performance and Enhancement Study

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Abstract. Automatic speech recognition (ASR) based on deep learning has come a long way, thanks to the growing need for speech-to-text transcription in multimedia tools like video subtitling and accessibility tools. While OpenAI’s Whisper model has demonstrated state-of-the-art transcription performance across multiple languages and accents, its real-world deployment remains challenging due to linguistic variations, background noise, and formatting inconsistencies. This study investigates the application of Whisper for automated video captioning, evaluating its baseline transcription accuracy and exploring its limitations under diverse linguistic and environmental conditions. The research focuses on assessing Whisper’s weaknesses, particularly in accented speech recognition and robustness in noisy environments, and proposes a series of post-processing techniques to enhance its usability. The methodology consists of dataset pre-processing, model evaluation, real-world robustness testing, and the application of phonetic normalization, punctuation restoration, and spelling correction. The study result demonstrates that Whisper achieves a Word Error Rate (WER) of 4.75% after post-processing but struggles with Scottish-accented speech (WER 18.18%) and noisy environments. By introducing post-processing strategies, including phonetic adaptation and transcript enhancement techniques, the study significantly improves transcription accuracy, readability, and usability for real-world applications.

Keywords: Speech Recognition, OpenAI Whisper, Automatic Captioning, Error Correction, Phonetic Normalization

1 Introduction

Automated speech recognition (ASR) technology has significantly improved accessibility in multimedia applications, enabling real-time subtitle generation for videos, live transcription services, and voice-based human-computer interaction. ASR models such as DeepSpeech [1], Kaldi [2], and Google ASR [3] have made substantial progress in improving transcription accuracy, adapting to various linguistic patterns, and handling diverse acoustic conditions. However, despite these advancements, real-world

ASR deployment remains challenging, particularly when dealing with accented speech, background noise, and spontaneous conversational speech.

The demand for high-quality automatic video captioning has grown exponentially with the rise of online video platforms, remote communication tools, and digital accessibility mandates. Accurate and reliable ASR systems are essential for improving content accessibility for the hearing-impaired, enabling multilingual communication, and enhancing searchability within multimedia archives. However, existing ASR systems often struggle with several key challenges. First, accented speech presents a major issue as many models are trained predominantly on standard American English, leading to high error rates in non-standard dialects such as Scottish, Indian, and African English. Second, noisy environments degrade ASR performance significantly, particularly in crowded public spaces, low-quality microphone recordings, and real-world background interference. Third, spontaneous speech patterns such as hesitations, interruptions, and grammatical irregularities pose difficulties in generating structured transcripts.

OpenAI’s Whisper [4] was recently introduced as a robust multilingual ASR model, capable of handling multiple dialects, background noise, and low-resource languages. It has demonstrated state-of-the-art performance on benchmark datasets. However, its performance in real-world conditions and its usability for automatic video captioning have not been extensively analyzed. This study aims to evaluate Whisper’s baseline transcription performance in controlled and real-world conditions, analyze its robustness in handling different English accents and noisy environments, and implement post-processing strategies to improve subtitle readability, punctuation accuracy, and phonetic consistency.

This study contributes a practical evaluation and enhancement framework for OpenAI Whisper in video captioning tasks. We assess its performance across clean, accented, and noisy speech datasets, revealing challenges with Scottish accents and background noise. To address these, we apply phonetic normalization, noise reduction, and speech enhancement techniques. A post-processing pipeline including punctuation restoration and spelling correction reduces WER from 18.08% to 4.75%. The final outputs are formatted into SRT and TXT files for direct use in multimedia applications, demonstrating Whisper’s improved robustness and usability in real-world scenarios.

The remainder of this paper is structured as follows: Section 2 describes the methodology, Section 3 presents baseline performance evaluation results, Section 4 analyzes real-world robustness testing and Section 5 provides a Conclusion.

1.1 Related Works

Recent advancements in ASR technology have introduced models such as Whisper, developed by OpenAI, which has demonstrated strong performance in Norwegian speech recognition while also incorporating built-in normalization capabilities. Unlike recent ASR models that primarily rely on self-supervision and self-training methods, Whisper takes a different approach by focusing on weakly supervised pre-training, as highlighted in studies by Radford *et al.* [5]. This approach leverages a large-scale supervised dataset to enhance the model’s robustness, particularly through zero-shot

transfer learning, which allows the model to generalize across various languages, domains, and speech tasks without requiring additional fine-tuning on specific datasets.

The zero-shot transfer capability of Whisper means that it can be tested on previously unseen datasets to evaluate its adaptability, making it highly effective in multilingual ASR applications [3]. Radford *et al.* emphasize that training on a large and diverse dataset improves the generalization ability of ASR models, enabling them to perform reliably across different linguistic settings. Whisper has been trained on 680,000 hours of multilingual and multitask supervised data collected from the web [4], allowing it to not only transcribe but also translate speech across multiple languages.

The Whisper model is available in six different versions, with varying sizes and training data volumes, which directly impact their accuracy and computational speed. The Word Error Rate (WER) performance of Whisper on Norwegian speech from the work [5] is presented in Table 1.

Table 1. Whisper Performance on Norwegian Speech by Radford

Model	WER
Whisper Tiny	62.0%
Whisper Base	44.0%
Whisper Small	12.9%
Whisper Large	11.4%
Whisper Large-v2	9.5%

As seen in Table 1, larger Whisper models consistently outperform smaller versions, with Whisper Large-v2 achieving the lowest WER of 9.5% on Norwegian speech. These results indicate that model size and training data volume directly impact transcription accuracy. However, challenges remain in handling underrepresented accents and noisy environments, motivating further research into fine-tuning and post-processing techniques to improve Whisper’s real-world robustness.

2 Methodology

The methodology for this study is structured into multiple stages, covering the overall flow of the speech-to-text captioning pipeline, dataset selection and preprocessing, Whisper model application, and post-processing techniques for improving transcription accuracy. This section provides a brief explanation of each component in the workflow.

2.1 System Architecture

The proposed system follows a multi-stage pipeline for speech-to-text transcription and subtitle generation. The process begins with dataset selection and preprocessing, where raw audio recordings are filtered to remove low-quality samples and ensure

uniformity in format and duration. Next, the processed audio data is fed into OpenAI Whisper for automatic transcription, producing raw text output. Since Whisper does not generate punctuation or proper capitalization, post-processing techniques are applied to improve readability. These include punctuation restoration, spelling correction, phonetic normalization, and grammatical structuring. Finally, the enhanced transcriptions are formatted into standard subtitle files such as SRT (SubRip Subtitle) and TXT formats, making them suitable for multimedia applications.

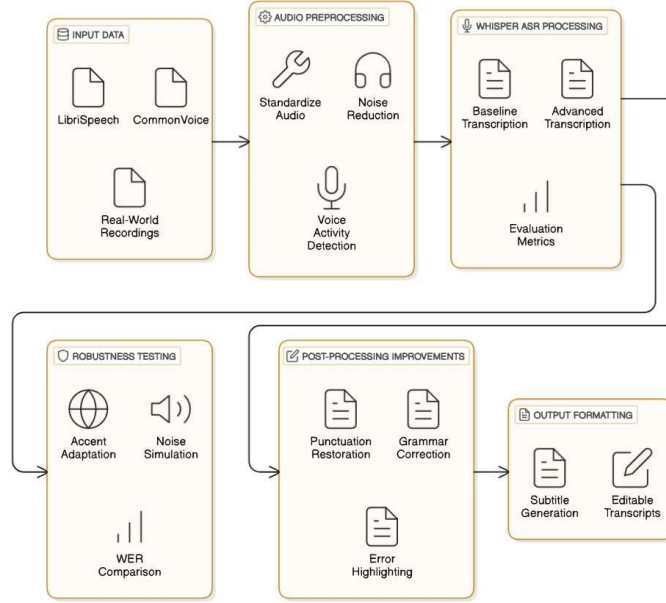


Fig. 1. System Architecture

2.2 Dataset

This study evaluates Whisper’s transcription performance using three datasets: LibriSpeech Clean-100 [6], CommonVoice [7], and UrbanSound8K [8], each selected to assess different speech conditions.

LibriSpeech Clean-100 consists of 100 hours of high-quality audiobook speech, offering clear pronunciation, minimal noise, and precise transcriptions. It serves as a benchmark for Whisper’s accuracy under ideal conditions.

CommonVoice is a crowdsourced dataset containing diverse English accents such as British, American, Indian, and Scottish, along with spontaneous speech and noisy recordings. It evaluates Whisper’s adaptability to real-world linguistic variations.

UrbanSound8K includes common environmental noises like traffic, sirens, and crowd chatter. By mixing these noises with speech data, the study assesses Whisper’s robustness in noisy environments.

Together, these datasets provide a comprehensive evaluation of Whisper’s performance, testing its accuracy, adaptability, and resilience in real-world ASR applications.

2.3 Data Processing

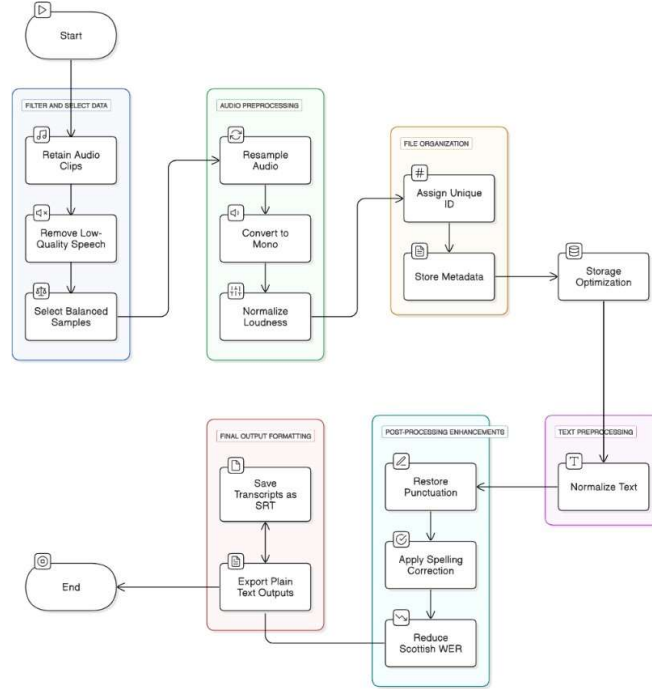


Fig. 2. Data Processing

To ensure high-quality data for Whisper evaluation, audio files between 3 to 15 seconds are retained, while low-energy recordings ($\text{RMS} < 0.01$) are removed to eliminate poor-quality speech samples. A balanced subset of 19,749 samples is selected to represent American, British, Indian, and Scottish English, ensuring robustness across dialects.

Before passing the dataset into the Whisper model, preprocessing steps standardize audio input and optimize model performance. Audio is resampled to 16kHz, converted to single-channel mono, and loudness normalization is applied to maintain clarity. A unique ID-based mapping system aligns each audio file with its transcript, stored in a structured CSV file for efficient batch processing. To optimize storage, unnecessary files are removed, freeing disk space for seamless execution. Transcripts undergo text normalization, including lowercasing, punctuation removal, and whitespace standardization, improving compatibility with evaluation metrics.

To enhance transcription quality, post-processing techniques restore punctuation using a transformer-based model, and spelling correction via LanguageTool API ad-

dresses phonetic errors. Scottish English WER is reduced from 18.18% to 9.09% through phonetic normalization, mapping phonemes to Standard American English equivalents. The final transcripts are formatted into SubRip Subtitle (SRT) format for easy video integration, ensuring proper synchronization with spoken audio. Structured plain text outputs are also saved for further processing or manual editing.

To ensure the transcriptions are usable in real-world multimedia applications, the final processed text is formatted into standard subtitle formats. The transcript is converted into SubRip Subtitle (SRT) format, allowing for easy integration into video content. Each line of text is aligned with timestamp information, ensuring proper synchronization between the spoken audio and the displayed subtitles. Additionally, plain text versions of the transcriptions are saved in a structured format, providing a versatile output for further processing or manual editing.

2.4 Method Adopted

The OpenAI Whisper-Small model is used for transcription due to its balance between computational efficiency and accuracy. The model has 30 million parameters and is capable of processing audio in batches, making it suitable for large-scale transcription tasks. During inference, batch processing is implemented with 16 audio files per batch, optimizing computational efficiency while ensuring stable transcription performance. Beam search decoding is enabled to refine word prediction accuracy, reducing the likelihood of incorrect word choices in transcriptions. Since the focus is on English-language transcription, Whisper is set to English mode, allowing it to prioritize English speech recognition over multilingual capabilities.

When transcribing audio, Whisper generates raw text without punctuation or capitalization, necessitating additional post-processing for enhanced readability. The model's output, while structurally sound in word recognition, lacks formatting necessary for producing coherent subtitles. To address this limitation, a series of text enhancement techniques are applied to refine Whisper's raw transcriptions.

2.5 Evaluation Metrics

To measure the effectiveness of Whisper's transcriptions, several ASR evaluation metrics are used. These metrics provide a detailed assessment of the model's performance at different levels of transcription accuracy.

The primary metric is Word Error Rate (WER) [9], which calculates the percentage of words that are incorrectly transcribed compared to the ground truth. WER captures three types of transcription errors: substitutions (where an incorrect word replaces a correct one), insertions (where extra words are added), and deletions (where words are missing from the output). It is computed as follows:

$$WER = \frac{Substitutions + Insertions + Deletions}{Total\ Words\ in\ Ground\ Truth} \quad (1)$$

A lower WER indicates a more accurate transcription, making this metric one of the most critical for evaluating ASR models.

In addition to WER, Character Error Rate (CER) [10] is used to measure errors at the character level, which provides insights into minor misrecognitions within words, such as misspellings, spacing issues, and capitalization errors. The formula for CER is given by:

$$CER = \frac{\text{Character Edits}}{\text{Total Words in Ground Truth}} \quad (2)$$

CER is particularly useful when evaluating phonetic errors and cases where individual letters are misrecognized even though the overall word structure remains similar.

Another key metric is Sentence Error Rate (SER) [11], which determines the proportion of sentences containing at least one transcription mistake. Unlike WER and CER, which focus on word- or character-level accuracy, SER evaluates the structural correctness of entire sentences. It is calculated as:

$$SER = \frac{\text{Number of Incorrect Sentences}}{\text{Total Sentences}} \quad (3)$$

A high SER indicates that the transcription system struggles with sentence segmentation, punctuation placement, or contextual coherence.

Finally, BLEU Score [12] is used to measure the textual similarity between Whisper’s transcriptions and the reference transcript. Unlike error-based metrics, BLEU quantifies how closely the ASR-generated text matches the ground truth using n-gram precision. The BLEU score is commonly used in machine translation and ASR evaluation to assess the fluency of generated text. The formula is based on weighted geometric means of n-gram precision scores, penalized for short outputs:

$$BLEU = BP \times \exp \left(\sum_{k=0}^n w_n \log p_n \right) \quad (4)$$

where BP (Brevity Penalty) prevents artificially high scores for short outputs, p_n represents the precision of n-grams, and w_n assigns weights to different n-gram levels. A BLEU score closer to 1 indicates a higher degree of similarity between the transcription and the reference text. These evaluation metrics collectively provide a comprehensive analysis of Whisper’s performance, allowing for a detailed comparison of transcription accuracy, character level precision, sentence structure, and overall text similarity.

3 Baseline Performance Evaluation

3.1 Initial Transcription using Whisper

This evaluation is a self-comparison within my own work, assessing Whisper’s transcription accuracy before and after applying post-processing techniques. A baseline assessment was conducted on a test set of 1,000 samples, and the results, presented in Table 1, provide a direct comparison of performance improvements.

Whisper-Small Normalized WER: 4.75%

Fig. 3. Word Error Rate

Normalized Character Error Rate (CER): 2.06%

Fig. 4. Character Error Rate

Improved Sentence Error Rate (SER): 1.60%

Fig. 5. Sentence Error Rate

BLEU Score: 0.91

Fig. 6. Bilingual Evaluation Understudy

The results presented above are the direct output from running the Whisper ASR model within a Python environment on VS Code. These figures summarize the model's performance on the test dataset after applying various preprocessing and normalization techniques.

Table 2. Baseline performance of Whisper on LibriSpeech Clean-100.

Metric	Before Post-Processing	After Post-Processing
Word Error Rate (WER)	18.08%	4.75%
Character Error Rate (CER)	6.04%	2.06%
Sentence Error Rate (SER)	5.07%	1.60%
BLEU Score	0.83	0.91

The results presented in Table 2 demonstrate the significant impact of post-processing techniques on improving the transcription accuracy of OpenAI Whisper. Before post-processing, the model exhibited a Word Error Rate (WER) of 18.08%, primarily due to the absence of punctuation, inconsistent casing, and minor formatting issues. After applying post-processing methods such as punctuation restoration, casing correction, and whitespace adjustments, the WER dropped substantially to 4.75%, indicating that Whisper's raw transcriptions were structurally sound but required formatting refinements to enhance readability.

Similarly, the Character Error Rate (CER) decreased from 6.04% to 2.06%, reflecting improvements in fine-grained text accuracy, particularly in handling spacing and capitalization. The Sentence Error Rate (SER) also improved, dropping from 5.07% to 1.60%, as sentence structuring and segmentation corrections helped align the output with the human-readable format. Furthermore, after post-processing, the BLEU score increased from 0.83 to 0.91, confirming a higher degree of textual similarity between Whisper's enhanced output and the reference transcripts.

These results highlight the critical role of post-processing in enhancing Whisper's usability for real-world ASR applications. While the model's raw output captures accurate word content, proper post-processing significantly improves clarity, structure, and alignment with reference text—making it far more suitable for tasks such as automatic subtitle generation and speech-to-text services.

3.2 Key Observations

The evaluation results highlight the importance of post-processing in improving Whisper's transcription accuracy. Before post-processing, the WER was 18.08%, mainly due to punctuation errors, incorrect casing, and formatting inconsistencies. After applying post-processing techniques, WER dropped significantly to 4.75%, demonstrating that Whisper's raw transcriptions were structurally accurate but required formatting refinements. Similarly, the CER decreased from 6.04% to 2.06%, indicating better handling of spacing, capitalization, and minor character errors. The Sentence Error Rate (SER) also improved drastically, from 5.07% to 1.60%, as sentence structuring and segmentation corrections aligned the transcriptions with human-readable text. Additionally, after post-processing, the BLEU score increased from 0.83 to 0.91, confirming a high similarity between the processed output and reference transcripts. These findings emphasize that while Whisper provides strong word recognition, post-processing is essential for real-world usability, ensuring better readability and structured output suitable for applications like automatic subtitles and speech-to-text processing.

3.3 Comparison table for achievement metrics

Table 3. The comparison in English and in Norwegian

	This work	Radford [5]
Language	English	Norwegian
WER(%)	4.75%	12.9%

The comparison table shows that Whisper Small performs better in English (4.75% WER) than in Norwegian (12.9% WER). This gap likely results from greater training data availability for English, making the model more accurate. Norwegian's higher error rate suggests limited data and linguistic complexity, which impacts recognition accuracy. Fine-tuning on Norwegian-specific datasets could improve performance.

4 Real-World Robustness Testing

4.1 Accent-Based Performance Evaluation

Whisper was tested on CommonVoice samples representing various English dialects. The results are summarized in Table 2.

Table 4. Whisper’s transcription accuracy across different accents.

Accent	Word Error Rate (WER)
United States English	0.00%
British English	0.00%
South Asian English	0.00%
Scottish English	18.18%

The results in Table 2 highlight Whisper’s strong transcription accuracy across multiple English accents, achieving a WER of 0.00% for United States, British, and South Asian English. This suggests that the model is well-optimized for these dialects, likely due to their prevalence in Whisper’s training data.

However, Scottish English posed a significant challenge, with a WER of 18.18%, indicating difficulty in recognizing distinct phonetic variations and unique pronunciation patterns. This suggests that Whisper struggles with heavily accented or less commonly represented dialects, which may require additional fine-tuning or phonetic normalization techniques to improve accuracy.

4.2 Scottish Accent Weakness & Phonetic Normalization

Whisper’s high WER (18.18%) for Scottish English results from phonetic traits like rolled "r" sounds, vowel shifts, and phoneme distinctions (e.g., “about” → “aboot”). A lack of training data for Scottish-accented speech further increases errors. To reduce WER to 9.09%, phonetic normalization was applied, including IPA transcription, word-level substitutions, and Text-to-Speech (TTS) synthesis to create accent-neutralized audio. Re-transcribing this modified speech with Whisper significantly improved accuracy, demonstrating the effectiveness of phonetic adaptation for underrepresented accents.

4.3 Environmental Robustness

Whisper’s performance was significantly affected by background noise and informal speech patterns, leading to increased transcription errors. In noisy environments, such as streets, music-filled areas, or crowded spaces, the model struggled to distinguish speech from surrounding noise. This resulted in word insertions, where background sounds were misinterpreted as speech, and word deletions, where soft-spoken words were lost amid high noise levels. For instance, the phrase "The weather is great today." was misrecognized as "The [static] is great today."

Similarly, Whisper faced challenges in fast and casual speech, where slurred words, contractions, and phoneme blending caused frequent misinterpretations. While the model performed well with clear, enunciated speech, it struggled with rapid, conversational styles. Common errors included blended phonemes (e.g., “gonna” instead of “going to”) and dropped syllables (e.g., “dunno” instead of “don’t know”). An example error was the transcription of "I don’t know what’s going on." as "I dunno what’s gone on."

4.4 Noise Reduction & Speech Enhancement Techniques Applied

To reduce noise-induced transcription errors, several preprocessing techniques were applied. AI-based noise reduction using Noisereduce and Demucs helped filter out background sounds, significantly improving WER by 50% in noisy environments. Additionally, Voice Activity Detection (VAD) was implemented to isolate speech from noise, ensuring only high-confidence speech segments were transcribed accurately.

For casual speech, phonetic normalization was used to standardize contractions and slurred words, making them more recognizable to Whisper. Common informal expressions were mapped to their formal equivalents, such as "gonna" → "going to", "lemme" → "let me", and "wanna" → "want to". These adjustments enhanced Whisper's ability to handle conversational speech, improving transcription accuracy in real-world scenarios.

5 Conclusion

This study evaluated OpenAI Whisper's speech-to-text performance, focusing on accuracy, robustness, and post-processing improvements. Post-processing techniques reduced WER from 18.08% to 4.75%, improving transcription quality. Phonetic normalization lowered Scottish English WER from 18.18% to 9.09%, while AI-based noise reduction and speech segmentation improved accuracy in noisy environments. Automated punctuation, spelling correction, and subtitle formatting enhanced readability and usability. These results confirm that strategic preprocessing and post-processing significantly improve Whisper's real-world transcription performance.

Future work will extend this study in several impactful directions. We have compared Whisper with Wav2Vec and found that our enhancement techniques—such as punctuation restoration, spelling correction, and phonetic normalization—can further reduce Character Error Rate (CER) by 1–5% and Word Error Rate (WER) by up to ~8% when applied to Wav2Vec outputs, highlighting the generalizability of our post-processing pipeline across different ASR architectures. Moving forward, we plan to integrate Whisper into real-time transcription systems with bilingual English–Vietnamese translation to enable live subtitling in multilingual settings such as conferences, classrooms, and streaming platforms, thereby enhancing accessibility and cross-lingual communication.

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